Project Proposal: Exploring transfer learning

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Project Group 33

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1. Whether you will complete a custom, default or default + extension project. Default + extension project.

   We will complete a default + extension project based on default project 1: explore transfer learning.

2. A brief description of the problem that you will work on and how you will try to solve it. Reference to at least one paper that provides specific inspiration as regards the problem statement and/or proposed approach to tackling the problem.

   We will work on Default Project 1 as described in the description. We will explore transfer learning by downloading a pre-trained network and fine-tuning it to recognize breed of cat or dog. We will explore different ways and circumstances of fine-tuning.

   We will first replace the final layer of a pre-trained ResNet34 to solve a dog-vs-cat sanity check. We will then adapt the model to the 37-class breed classification task and compare linear probing, partial fine-tuning, full fine-tuning, and gradual unfreezing. We will also study limited labeled data and class imbalance.

   We will also implement extensions to our network. The first extension will be to investigate **semi-supervised learning** using limited data. We will examine if unlabeled training samples can help recover some of the performance lost when only a small fraction of the training data is labeled. We plan to explore one approach, most likely pseudo-labeling inspired by Lee's *Pseudo-Label : The Simple and Efficient Semi-Supervised Learning Method for Deep Neural Networks* [1], and compare to standard supervised fine-tuning with the same amount of labeled data.

   The second extension, will be to compare **LoRA fine-tuning** with standard full fine-tuning, inspired by Hu et al.'s *LoRA: Low-Rank Adaptation of Large Language Models* [2]. This extension will only be implemented if the first extension is not deep enough and if time permits. The aim will be to investigate whether parameter-efficient fine tuning can achieve competitive performance while updating fewer trainable parameters.

3. The data that you will use for training, validation and testing.
   We will use the `Oxford-IIIT Pet Dataset` for training, testing and validation.

4. The deep learning software package(s) you will use.

   We will primarily use PyTorch and matplotlib for plotting and visualizations.

5. How much of the software implementation your group will write and to what extent will rely on open-source implementations.

   We will implement the full experimental pipeline ourselves, including data preprocessing, train/validation/test splitting, linear probing, partial and full fine-tuning, gradual unfreezing, limited-data experiments, imbalanced-data generation, evaluation, and visualization. We will use an open-source pretrained ResNet34 model as the starting point for transfer learning, since training a strong visual backbone from scratch is not the goal of this project. The main contribution of our project is not the pretrained backbone itself, but how we adapt and compare different transfer learning strategies on top of it.

6. The baselines you intend to use.

   We intend to use the following baselines: linear probing, partial fine-tuning of the last layers, gradual unfreezing, fine-tuning with limited data and imbalanced classes. For the semi-supervised extension, the main baseline will be standard supervised fine tuning using the same fraction of labeled data. For the LoRA extension, the main baseline will be full fine tuning of the same pretrained model.

7. How will you evaluate your results?

   The main evaluation metrics will be: classification accuracy, confusion matrices to analyze which breeds are most frequently confused with each other, training time and computational costs, performance as a function of amount of labeled data, and number of trainable parameters. We will also evaluate the effect of regularization and data augmentation such as flips, rotations, crops and small size scaling. The per-class evaluation metrics will more specifically be: per-class F1, per-class precision and recall.

8. The grade your project group is aiming for in the range A-E.

   We are aiming for **A**.

9. Which achieved milestones during your project should be reached as you work towards your grade target.

   (a) range E milestones:
   - Download and fine-tune final layer of pre-trained network for sanity check
   - Train and evaluate a 37-class breed classifier
   - Compare linear probing, partial fine-tuning, full fine-tuning, and gradual unfreezing
   - Study performance with limited labeled data
   - Construct and evaluate an imbalanced training setting

   (b) range A-B milestones:
   - Investigate semi-supervised learning, most likely pseudo-labeling
   - Compare semi-supervised learning to supervised fine-tuning with the same amount of labeled data
   - Implement and compare LoRA fine-tuning to full fine-tuning
   - Perform at least one ablation for LoRA, such as varying the insertion location.
   - Analyze in which settings semi-supervised learning and LoRA are most beneficial.

10. Specify for each group member the skills/knowledge w.r.t. deep learning "theory" and practice they aim to acquire.

    Alexander: Have taken two courses in machine learning covering small parts of deep learning. DD1418 and DD2421. Since this is the first course in Deep Learning, I aim to learn more and get some real practical experience of the topic rather than just understanding the theory. Hopefully, this will help me generalize the results and technique in order to be able to conduct other Deep Learning problems in my career.

    Emma: My goal is to apply my more theoretical knowledge into practical skills. I have previously done theory courses in machine learning, with not much real-world application. In this project I aim to learn how to apply deep learning libraries and figure out ow to adapt pre-trained networks to solve new problems.

    Joachim: I have studied machine learning, AI, and foundational mathematics courses. I used machine learning in my bachelor's thesis. I aim to learn more about how to improve and evaluate networks and how different configurations of data affect performance.

References

[1] Dong-Hyun Lee. *Pseudo-Label: The Simple and Efficient Semi-Supervised Learning Method for Deep Neural Networks*. ICML Workshop, 2013.

[2] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. *LoRA: Low-Rank Adaptation of Large Language Models*. arXiv preprint arXiv:2106.09685, 2021.